o6SG

Response: Sensitivity Analysis of WePE Hyperparameters

We thank the reviewer for this targeted question. To provide direct empirical evidence, we conducted a controlled sensitivity analysis sweeping four groups of hyperparameters on a 14×14 patch grid (224×224 images, patch size 16). For each configuration we report three metrics: (i) the Pearson correlation ρ between normalised patch distance and positional-encoding dissimilarity ($1 - \cos \mathrm{sim}$) over all $\binom{196}{2} = 19{,}110$ patch pairs, which quantifies how well the distance-decay property is preserved; (ii) the proxy training loss after 30 gradient steps on a depth-6 ViT-T, which probes training stability; and (iii) feature statistics (mean absolute value, saturation fraction $|f| > 0.99$, and near-zero fraction $|f| < 0.01$).

Table 1: Sensitivity to the pole-neighbourhood multiplier κ.

κ	ρ	Proxy loss	Mean$\|f\|$	Std	Sat.%	≈0%
1.0	0.6253	4.508	0.1063	0.225	1.91	11.61
5.0	0.6253	4.454	0.1063	0.225	1.91	11.61
15.0	**0.6253**	**4.471**	**0.1063**	**0.225**	**1.91**	**11.61**
30.0	0.6253	4.515	0.1063	0.225	1.91	11.61
60.0	0.6253	4.531	0.1063	0.225	1.91	11.61
120.0	0.6253	4.496	0.1063	0.225	1.91	11.61

(A) Pole-neighbourhood threshold κ ($|z| < \kappa\varepsilon$ triggers $C_{\mathbf{large}}$ substitution). All three metrics are *identical* across the entire range $\kappa \in [1, 120]$ (CV(ρ) = 0.000, CV(loss) = 0.006). The reason is structural: for the lemniscatic square lattice with $\omega_1 \approx 2.622$, the normalised patch coordinates $(u, v) \in (0, 1)^2$ are mapped to $|z| \geq 0.5\omega_1/H \approx 0.094$, which is already $\sim 10^7\varepsilon$ away from the origin ($\varepsilon = 10^{-8}$). Consequently, *no patch coordinate ever falls within the clipping radius for any tested* κ, and the κ parameter is effectively inactive under standard coordinate normalisation. This confirms that Eq. (7) of the paper is robust to the choice of κ over more than two orders of magnitude.

Table 2: Sensitivity to the pole substitute C_{large}.

C_{large}	ρ	Proxy loss	Sat.%
5×10^1	0.6253	4.459	1.91
2×10^2	0.6253	4.500	1.91
5×10^2	0.6253	4.476	1.91
2×10^3	0.6253	4.459	1.91
5×10^3	0.6253	4.488	1.91
1×10^5	0.6253	4.519	1.91

(B) Pole substitute value $C_{\mathbf{large}}$. ρ and the saturation fraction are *completely invariant* to C_{large} across four orders of magnitude (CV(ρ) = 0.000, CV(loss) = 0.005). Since no patch coordinate activates the clipping condition (see above), the value assigned at the pole is never used in practice; the tanh compression further ensures that even if an extreme value were occasionally assigned, it would be mapped to ± 1 before projection.

Table 3: Sensitivity to the lattice scaling factor $\alpha_u = \alpha_v$.

$\alpha_u = \alpha_v$	ρ	Proxy loss	Mean$\|f\|$
0.20	0.613	4.464	0.323
0.40	**0.625**	**4.488**	**0.106**
0.60	0.620	4.473	0.053
0.80	0.513	4.469	0.040
1.00	0.347	4.556	0.073
1.20	0.089	4.478	0.100

(C) Lattice scaling factors $\alpha_u = \alpha_v$. The lattice scaling factors are the *only* hyperparameters that meaningfully affect ρ (CV = 0.418): values in $\alpha \in [0.2, 0.6]$ produce strong distance decay ($\rho \in [0.61, 0.63]$), whereas larger values push patch coordinates deeper into higher-frequency oscillation regions of $\wp(z)$, reducing the monotonicity of the decay. Importantly, this sensitivity is *by design*: α_u and α_v are the primary geometric control of the encoding, analogous to choosing frequency bands in sinusoidal PE. In all our experiments $\alpha_u = \alpha_v = 0.4$ is set as the default, and the learnable parameter ω_3' (Eq. 2) allows the aspect ratio to adapt during training. The proxy loss varies only within 1% ($\Delta = 0.007$) across the entire range, confirming that training stability is not sensitive to this choice even when ρ degrades at extreme values.

Table 4: Sensitivity to the tanh compression factor α_{scale}.

α_{scale}	ρ	Proxy loss	Sat.%	$\approx$0%
0.010	0.634	4.485	0.26	84.82
0.050	0.630	4.559	0.89	42.98
0.100	0.627	4.434	1.40	20.92
0.150	**0.625**	**4.442**	**1.91**	**11.61**
0.300	0.625	4.492	3.32	5.87
0.500	0.626	4.486	4.34	3.57

(D) tanh **compression factor α_{scale}.** ρ is essentially flat across the entire range (CV = 0.005, $\Delta\rho < 0.009$), confirming that the distance-decay property is stable. The proxy loss fluctuates within ± 0.06 (CV = 0.009), within the noise of 30 gradient steps. The feature utilisation trade-off is visible: very small α_{scale} wastes representational capacity (84.8% of features near zero at $\alpha = 0.01$), while very large values increase saturation. The default $\alpha_{\text{scale}} = 0.15$ balances these two effects. In practice α_{scale} is implemented as a learnable parameter via `softplus`, so it adapts automatically during training.

Overall conclusion. Table 5 summarises the coefficient of variation (CV) across all sweeps.

Two hyperparameters (κ, C_{large}) are *structurally inert* under standard coordinate normalisation and can be set to any reasonable value without consequence. The tanh compression factor α_{scale} has negligible effect on ρ and loss (CV < 0.01) and is in any case learnable. Only the lattice scaling

Table 5: Coefficient of variation (CV = std/mean) of key metrics across each hyperparameter sweep. CV < 0.01 indicates negligible sensitivity.

Sweep	CV(ρ)	CV(loss)	CV(mean$\|f\|$)
A: threshold κ	0.0000	0.0059	0.0000
B: C_{large}	0.0000	0.0049	0.0000
C: $\alpha_u = \alpha_v$	0.4175	0.0070	0.8230
D: α_{scale}	0.0049	0.0091	0.6610

factors α_u, α_v require care: the interval $[0.2, 0.6]$ consistently yields $\rho \geq 0.61$, and the learnable ω_3' automatically compensates for aspect-ratio variation within this range during training. These results confirm that WePE is robust and genuinely plug-and-play across all hyperparameters except the spatial frequency range, which is controlled by the single learnable parameter ω_3' and the fixed default $\alpha = 0.4$.

Response to Computational Benchmark Request

We thank the reviewers for requesting direct empirical evidence of computational efficiency. We have implemented a controlled benchmark measuring inference latency, training-step time, and peak GPU memory across three architectures (ViT-T/B/L) and two resolutions (224×224 and 384×384), comparing WePE-LUT against APE, 2D-Sinusoidal PE (2D-Sin), RoPE-Mixed, and FoPE. All measurements were taken on an NVIDIA RTX 4090 (24 GB, CUDA 12.1, PyTorch 2.2.0) with batch size 32 for inference and 16 for training, using 20 warm-up and 100 timed iterations; GPU-side timing was obtained via `torch.cuda.Event` to eliminate host-synchronisation overhead. Results are reported in Tables 6 and 7.

WePE-LUT vs. direct lattice summation. Without the look-up table (LUT), evaluating $\wp(z)$ via truncated lattice summation ("WePE-Direct") incurs inference latencies of 532–540 ms on ViT-T, roughly **90**× slower than APE. This confirms that direct evaluation is impractical for online use. The LUT pre-computation eliminates this cost entirely: once trained, the optimal elliptic-function parameters are fixed, a $256 \times 256 \times 4$ look-up table is stored as a model buffer, and patch positional features are retrieved at forward time via hardware-accelerated bilinear interpolation (`F.grid_sample`), as described in Appendix F.

WePE-LUT adds negligible latency overhead. The inference latency of WePE-LUT matches that of all baselines to within measurement noise across every tested configuration. On ViT-T at 224×224, WePE-LUT achieves 6.35 ms versus 5.83 ms for APE (+0.52 ms, +8.9%); at 384×384 the gap is 18.33 ms versus 18.23 ms (+0.10 ms, +0.5%). On ViT-B and ViT-L the differences are similarly within 0.5%, consistent with the $\mathcal{O}(N \cdot d)$ time complexity shared by all methods once the LUT is pre-computed. Training-step times are equally comparable: on ViT-B at 224×224, WePE-LUT requires 49.41 ms versus 48.38 ms for APE (+2.1%).

Memory overhead. WePE-LUT does incur a higher peak memory footprint than APE or 2D-Sin, primarily because the lightweight MLP projection head (mapping $R^4 \to R^d$ with a `LayerNorm`) introduces $\mathcal{O}(d)$ additional parameters and its associated activation tensors. The static 256×256×4 LUT buffer contributes a fixed 1 MB regardless of architecture or resolution, and is negligible. As shown in Table 6, the absolute memory difference between WePE-LUT and APE is 227 MB on ViT-T, 2,693 MB on ViT-B, and 9,332 MB on ViT-L at 224×224. We note that the ViT-L figures reflect activation memory dominated by 24 transformer layers under full back-propagation with batch size 16; the PE module itself accounts for a small fraction of this total. The ViT-L@384 configuration exceeds the available VRAM (24 GB) for all methods during training, which is a hardware constraint rather than a WePE-specific limitation; inference completes successfully for all methods at that setting.

Summary. These results confirm the central claim of Section 2.2 and Appendix F: once the LUT is pre-computed, WePE introduces *no measurable latency overhead* relative to simpler baselines, at the cost of a moderate memory increase attributable to the projection head. The time complexity of online WePE-LUT is $\mathcal{O}(N \cdot d)$, identical to APE and 2D-Sin, as established theoretically in Appendix F and now confirmed empirically across three architectures and two resolutions.

Table 6: Benchmark results at resolution 224 × 224. Inference: batch 32, 100 iterations. Training step: batch 16, forward+backward+optimizer, 20 iterations. Peak GPU memory includes activations and gradients. WePE-Direct is reported only for ViT-T (too slow for larger models).

Arch	PE Method	Inf. (ms)	Inf. Mem (MB)	Trn. (ms)	Trn. Mem (MB)	PE params
6*ViT-T	APE	5.83	113	22.57	669	38,016
	2D-Sin	6.01	165	23.07	713	192
	RoPE-Mix	6.06	209	21.82	756	12,480
	FoPE	6.28	252	22.21	800	3,464
	WePE-Direct	532.56	295	1476.41	857	1,731
	WePE-LUT	**6.35**	**340**	**22.86**	**888**	**58,087**
5*ViT-B	APE	27.21	612	48.38	3,340	152,064
	2D-Sin	27.28	1,283	48.52	4,025	768
	RoPE-Mix	27.25	1,958	48.84	4,703	49,920
	FoPE	27.30	2,628	48.90	5,373	38,424
	WePE-LUT	**27.40**	**3,304**	**49.41**	**6,045**	**895,879**
5*ViT-L	APE	81.98	1,521	150.99	9,604	202,752
	2D-Sin	82.10	3,851	151.22	11,932	1,024
	RoPE-Mix	82.13	6,183	151.16	14,263	66,560
	FoPE	82.08	8,515	151.65	16,596	67,616
	WePE-LUT	**82.23**	**10,853**	**151.71**	**18,935**	**1,587,719**

Response to Question 1 (Appendix H.4)

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Table 7: Benchmark results at resolution 384×384. ViT-L training exceeds 24 GB VRAM for all methods (OOM); inference results are reported. WePE-LUT latency remains within 0.5% of APE at every scale where a valid comparison exists.

Arch	PE Method	Inf. (ms)	Inf. Mem (MB)	Trn. (ms)	Trn. Mem (MB)	PE params
6*ViT-T	APE	18.23	419	27.10	2,367	110,976
	2D-Sin	18.24	463	26.93	2,410	192
	RoPE-Mix	18.23	507	27.37	2,454	12,480
	FoPE	18.30	550	27.45	2,497	3,464
	WePE-Direct	540.48	594	1491.96	2,576	1,731
	WePE-LUT	**18.33**	**638**	**27.39**	**2,585**	**58,087**
5*ViT-B	APE	98.57	1,270	171.36	10,082	443,904
	2D-Sin	98.62	1,936	170.54	10,751	768
	RoPE-Mix	98.56	2,606	170.36	11,403	49,920
	FoPE	98.66	3,270	170.77	12,068	38,424
	WePE-LUT	**98.71**	**3,942**	**170.28**	**12,740**	**895,879**
5*ViT-L	APE	294.52	2,389	OOM		591,872
	2D-Sin	294.65	3,553	OOM		1,024
	RoPE-Mix	294.56	4,713	OOM		66,560
	FoPE	294.68	5,875	OOM		67,616
	WePE-LUT	**294.93**	**7,042**	OOM		**1,587,719**

Regimes where WePE provides a practically meaningful advantage over simpler 2D-preserving alternatives.

We thank the reviewer for this important question. We agree that the gap between WePE and 2D sinusoidal encoding in Appendix H.4 is small under that particular setting (ViT-Tiny on CIFAR-100, trained from scratch for 60 epochs). We intentionally included this comparison to make clear that, in a lightweight regime with limited model capacity, limited data, and short optimization, preserving the 2D structure itself already captures much of the available gain over flattened 1D encodings, so stronger geometric priors may not yet translate into a large absolute improvement.

Our claim is therefore not that WePE universally yields a large margin over simpler 2D-preserving alternatives, but rather that its additional mathematical structure becomes more practically meaningful in regimes where the model has sufficient capacity and supervision to exploit richer geometric information. In our current results, these regimes are mainly the following:

(1) Larger-scale training / longer optimization. The small gap in Appendix H.4 is observed in a deliberately constrained setting. When training is extended and/or the data scale increases, the advantage of WePE becomes more visible. For example, in Table 1 (120 epochs), WePE achieves higher top-1 accuracy than the other positional encodings reported there, and in Table 2 the advantage remains present across larger training fractions and also on ImageNet-1K. We view this as evidence that the extra geometric structure of WePE is more useful once the model has enough data and training time to benefit from it, whereas in short small-scale training a simpler 2D sinusoidal encoding is already a strong baseline.

(2) Tasks that benefit from explicit locality and distance-aware geometry. A key difference is that WePE is not only 2D-preserving, but also induces a structured locality prior through its analytically motivated distance-decay behavior. In the paper, this is reflected both theoretically and empirically: Figure 3 shows that WePE produces more spatially coherent attention patterns already at initialization, and the distance-decay analysis shows that nearby patches receive more similar positional representations. Such properties are expected to matter more in tasks where precise spatial relationships are central, rather than in small-scale image classification alone.

(3) Dense prediction and localization-sensitive settings. This is the regime where we believe the additional structure is most practically meaningful. In Appendix H.11, when integrating WePE into dense prediction pipelines, we observe improved attention–mask alignment in the zero-shot COCO analysis, as well as consistent gains after replacing only the positional encoding inside a ViT-based detection/segmentation backbone. These results suggest that the value of WePE is not merely that it preserves 2D indexing, but that its geometry-aware structure provides a stronger inductive bias for localization-sensitive tasks, where relative distance and spatial continuity are more critical than in a small CIFAR-scale benchmark.

(4) Resolution transfer / fine-tuning across different spatial layouts. Both 2D sinusoidal encodings and WePE are continuous, but WePE additionally uses a learnable lattice parameterization that can adapt its spatial geometry during fine-tuning. We believe this is particularly relevant when transferring pretrained models to different resolutions or aspect ratios. Consistent with this, Table 4 shows that WePE performs competitively and improves over the baseline on most VTAB-1k tasks, which suggests that the extra structure can be useful beyond the original pretraining grid.

In summary, we agree with the reviewer that Appendix H.4 shows only a modest gain over 2D sinusoidal encoding in a small-model, short-training classification regime. We therefore do not argue that the additional mathematical structure of WePE is uniformly necessary. Rather, our results indicate that it becomes practically meaningful in regimes involving (i) larger-scale or longer training, (ii) tasks requiring stronger locality and distance-aware geometry, (iii) dense prediction/localization, and (iv) transfer across resolutions. We will revise Appendix H.4 to make this scope clearer and to explicitly state that 2D sinusoidal encoding is a strong simple baseline in low-capacity settings, while WePE is most beneficial when richer geometric inductive bias is needed.

Response to the question on why the Weierstrass elliptic function was chosen over other doubly periodic functions.

We thank the reviewer for this thoughtful question. We agree that doubly periodicity alone does *not* uniquely single out the Weierstrass elliptic function, and we do *not* intend to claim in the paper that $\wp(z)$ is universally or absolutely superior to every other doubly periodic family (e.g., Jacobi elliptic functions) in all possible settings. A more precise statement is that Weierstrass elliptic functions provide a particularly natural and mathematically convenient *instantiation* of the design principles we seek for 2D positional encoding.

Our choice was guided by the following considerations.

(1) Canonical realization of doubly periodic geometry on a lattice / torus. A main conclusion of Appendix E is that, under standard desiderata such as translation equivariance, positive-definite realizability of the induced attention kernel, finite-dimensional approximation, and numerical stability, periodic (more precisely, torus-harmonic) structure is not ad hoc but structurally well motivated. In two spatial dimensions, this naturally points to a doubly periodic construction on a complex torus C/Λ. The Weierstrass elliptic function $\wp(z)$ is one of the most canonical realizations of this geometry: it is directly defined with respect to a lattice Λ, and its period parallelogram aligns naturally with the regular patch grid used in vision transformers. In this sense, the choice of $\wp$ is geometrically well matched to the image domain rather than being an arbitrary special function choice.

(2) Closed-form algebraic addition law for relative position modeling. A key reason we chose the Weierstrass family is not merely periodicity, but the availability of a classical algebraic addition formula. This matters because our goal is not only to encode absolute 2D position, but also to obtain a representation in which relative displacement can be derived in a structured way from absolute codes. As discussed in the manuscript, the addition law of $\wp$ allows pairwise positional interactions to be expressed algebraically from $\wp(z)$ and $\wp'(z)$, which provides a clean mathematical route to relative position modeling without introducing a separate bespoke relative-position module. This property was one of the main reasons for preferring the Weierstrass formulation as the backbone of WePE.

(3) The pair $(\wp(z), \wp'(z))$ gives a compact and expressive four-channel representation. In our construction, the real and imaginary parts of $\wp(z)$ and $\wp'(z)$ yield a compact four-dimensional feature that is both direction-aware and sensitive to local geometric variation. This is convenient architecturally: the representation is low-dimensional, continuous, and naturally compatible with a lightweight projection head. The derivative channel is also important in practice, as our ablations indicate that removing $\wp'(z)$ degrades performance. Thus, the Weierstrass formulation gives not only a doubly periodic scalar field, but also a natural companion derivative field that enriches the positional representation in a mathematically coherent way.

(4) Continuous evaluation and resolution-robust implementation. Another practical reason for our choice is that the Weierstrass construction is naturally continuous in the complex coordinate z. Once normalized patch coordinates are mapped into C, changing the image resolution only changes the evaluation points, not the encoding rule itself. This makes the construction well aligned with resolution transfer and lookup-table-based implementation. While other doubly periodic functions may also be continuous, the Weierstrass setting provides a particularly clean torus/lattice interpretation together with the addition-law structure above.

What about Jacobi elliptic functions? We agree that Jacobi elliptic functions are also important doubly periodic objects and are a valid comparison point conceptually. However, our goal in this paper was not to exhaustively benchmark all elliptic-function families, but to identify one principled, expressive, and implementation-friendly representative that matches the desired geometric properties. Weierstrass functions were attractive for this purpose because they are lattice-native, admit a symmetric torus formulation, and come with the classical $(\wp, \wp')$ algebraic machinery used throughout our theory. Compared with Jacobi elliptic functions, whose geometry is usually parameterized indirectly through the modulus k, the Weierstrass formulation is lattice-native: the underlying period lattice Λ appears explicitly in the definition. This makes the correspondence between the positional field and the 2D patch lattice more transparent in our setting. We therefore view $\wp$ as a *canonical and convenient choice*, rather than claiming that all alternatives are inferior in a theorem-level sense.

In summary, the Weierstrass elliptic function was chosen not because doubly periodicity uniquely forces it, but because it provides a particularly canonical, algebraically structured, and implementation-compatible realization of the doubly periodic positional geometry that our framework requires.

Response to the question on ultra-high resolutions / very small patch sizes.

We thank the reviewer for this important question. Our expectation is that WePE is *structurally well-suited* to higher resolutions, because it is defined on normalized continuous coordinates rather than on a fixed discrete positional table. In our formulation, patch centers are first mapped to $(u, v) \in [0, 1]^2$, and then evaluated by a continuous complex-valued positional field. Therefore, increasing the input resolution by itself does not invalidate the encoding: it simply samples the same underlying field more densely. In this sense, the theoretical behavior of WePE under ultra-high resolution is fundamentally different from that of discrete learned embeddings that must be interpolated from a fixed training grid.

From the theoretical side, the manuscript already makes two points relevant here. First, the positional map is resolution-invariant at the coordinate level, since the normalization in Eq. (1)–(2) keeps patch locations in a fixed compact domain regardless of how many patches are used. Second, for the direct lattice-sum construction, the truncation accuracy is governed by the chosen summation radius (or equivalently the desired numerical precision), rather than by the image resolution itself. Thus, moving to finer patch grids does not, by itself, require increasing the truncation order merely because the number of tokens is larger. What changes is the *sampling density* of the same field, not the mathematical definition of the field.

At the same time, we do not want to overstate this point: ultra-high resolutions and very small patch sizes do introduce practical challenges. The first is *computational*, since the token count grows rapidly and the overall transformer cost may dominate regardless of the positional encoding. The second is *geometric*: when patches become extremely small, the model may need to represent both very local detail and very long-range interactions within the same image, which places greater pressure on the choice of spatial scale parameters (e.g., α_u, α_v and the learnable half-period). The third is *numerical*: a denser spatial sampling means that more query points may lie close to regions where the raw elliptic field changes rapidly, so stabilization becomes more important even if the underlying field remains well-defined.

For this reason, our expected empirical behavior is the following. In the moderate high-resolution regime (the type already targeted by standard ViT fine-tuning), WePE should remain stable and should preserve its main advantage of geometry-aware, resolution-consistent positional information. In more extreme regimes with very small patches, we would still expect the method to function, but we would expect the *stabilized implementation* to become increasingly important. In particular, the fine-tuning surrogate described in Sec. 2.3 is designed exactly for settings where direct elliptic-function evaluation becomes unnecessarily fragile or expensive during optimization: it preserves the key geometric bias (strong local response plus directional periodicity) while replacing the singular raw field by a clamped and smooth approximation.

Accordingly, our answer is that no change to the *core idea* of WePE is required, but additional care in the *implementation* is advisable in the ultra-high-resolution / tiny-patch regime. The main stability mechanisms are already present in the paper:

1. **Adaptive feature compression.** In the direct formulation, each channel is passed through a scaled tanh, which prevents large magnitudes from propagating into the transformer.
2. **Near-pole clipping.** When the mapped coordinate is too close to a pole, the raw value is clipped to a large finite constant rather than evaluated directly.
3. **Controlled lattice summation.** The lattice sum is truncated with modulus-based ordering, which improves convergence and keeps the truncation error controlled.
4. **Stable surrogate for fine-tuning.** In Sec. 2.3, the radius is clamped by $r_{\text{safe}}(z) = \max\{r(z), \epsilon\}$ and the singular $1/\|z\|^2$ behavior is replaced by a finite, learnable surrogate term $1/(r_{\text{safe}}(z)^2 + \beta)$.
5. **LUT-based inference.** After training, a high-resolution lookup table with bilinear interpolation can be used; its approximation error is second-order in the grid spacing, so increasing the LUT resolution systematically reduces the interpolation error.

If one were to push WePE to especially extreme regimes (e.g., gigapixel-style inputs or unusually tiny patch sizes), the modifications we would recommend are therefore practical rather than conceptual:

1. increase the LUT resolution during inference so that interpolation error remains negligible under denser sampling;
2. use the stabilized surrogate/hybrid formulation during fine-tuning instead of repeated direct elliptic evaluation;
3. avoid pathological parameter settings that make the radial term too close to the original singular form (i.e., keep the stabilizing parameters such as β away from extremely small values);
4. if needed, retune or learn the spatial scale parameters so that the effective lattice frequency remains matched to the denser patch grid.

We also wish to be precise about the empirical evidence currently in the paper. Our experiments support the claim that WePE transfers well across resolution changes and that the proposed stabilization mechanisms are effective in practice. However, we have not run a dedicated benchmark specifically sweeping to *ultra*-high resolutions and extremely small patch sizes. Therefore, the strongest claim supported by the current manuscript is: *WePE is theoretically resolution-compatible and numerically stable under the proposed stabilization mechanisms, and these mechanisms should become increasingly important as the patch grid becomes denser.* We will clarify this scope in the revision.

In summary, our expectation is that WePE remains applicable in the ultra-high-resolution / small-patch regime because it is based on continuous normalized coordinates rather than fixed-grid embeddings. The main issue is not a breakdown of the theory, but the need for stronger numerical safeguards and efficient implementation. In practice, the existing stabilization tools in the paper (feature compression, clipping, safe-radius surrogate, and high-resolution LUTs) are precisely the mechanisms we would rely on to maintain stability in that regime.

Response to the question on whether the improvement comes primarily from double periodicity, or whether a non-pos-only method (e.g., semantic-aware PE) could also achieve SOTA.

We thank the reviewer for this important question. Our view is that the gains reported in this paper come *primarily from the positional inductive bias itself*, rather than from any hidden semantic adaptation, because our experiments were intentionally designed to isolate the effect of the positional encoding. In the main comparisons, the backbone, training recipe, and optimization settings are kept fixed, and the positional encoding is the only variable being changed. Therefore, the empirical improvements in our paper should be interpreted as evidence that a stronger *geometry-aware positional prior*—rather than additional semantic conditioning—is already sufficient to yield consistent gains.

More specifically, we do not attribute the improvement to "double periodicity alone" in a narrow sense. Rather, we attribute it to the *combination* of structural properties enabled by the doubly periodic Weierstrass construction:

1. a 2D lattice-native periodic geometry aligned with the patch grid,
2. continuous evaluation on normalized coordinates, which supports resolution transfer,
3. the algebraic addition-law structure, which makes relative spatial interactions more naturally expressible, and
4. a distance-sensitive locality bias, which encourages spatially coherent attention.

In other words, periodicity is a key ingredient, but the practical advantage of WePE comes from the full package of geometric structure provided by the Weierstrass formulation, not merely from periodicity in isolation.

At the same time, we fully agree with the reviewer that *non-pos-only* methods, including semantic-aware or content-adaptive positional mechanisms, may also achieve very strong results, potentially even state-of-the-art in specific tasks. However, such methods address a somewhat different question from the one studied in this paper. Once semantic information, task supervision, or content-conditioned modulation is injected into the positional representation, the resulting performance gain can no longer be attributed purely to positional geometry. Our goal here was deliberately narrower: to test whether a mathematically principled positional encoding *alone* can improve ViTs without introducing extra semantic branches, task-specific modules, or content-aware adaptation.

For this reason, we view semantic-aware PE and related non-pos-only approaches as largely *complementary* to our contribution rather than directly comparable alternatives. In fact, if a semantic-aware PE performs better, this would not contradict our claim; rather, it would suggest that geometry-aware positional structure and semantic adaptation can provide complementary benefits. The contribution of WePE is to show that even in the absence of explicit semantic conditioning, a better-structured positional field already improves attention organization, locality, and downstream performance. In principle, WePE could also be combined with semantic-aware mechanisms, which is an interesting direction for future work.

We also note that some of our auxiliary analyses reinforce this interpretation. For example, the zero-shot COCO attention-alignment experiment uses a frozen classifier without any task-specific segmentation supervision or semantic adaptation, yet WePE still yields better localization-aligned attention maps. This supports the view that the improvement is indeed coming from the geometry of the positional encoding itself, rather than from an auxiliary semantic mechanism. Similarly, our theoretical discussion motivates doubly periodic structure under translation-equivariant and positive-definite attention desiderata, suggesting that the positional prior is not merely a heuristic but a structurally meaningful component.

Therefore, the most precise answer is:

> The improvements in this paper are primarily due to a stronger *geometric positional prior* induced by the Weierstrass construction, of which double periodicity is an important part but not the only part. Semantic-aware or other non-pos-only methods may also achieve excellent or even stronger task performance, but they should be viewed as complementary rather than directly comparable, because they inject additional information beyond pure positional encoding.

In summary, our evidence suggests that the improvement is primarily from the richer positional structure introduced by WePE (not from hidden semantic adaptation), while semantic-aware PE represents a complementary class of methods that may further improve performance but addresses a broader design objective.

Response to the question on Figure 17 and the depth-wise expansion trajectory of the effective attention range.

We thank the reviewer for this insightful question. Our current evidence does *not* support claiming that the exact expansion trajectory in Figure 17 is a universal, dataset-invariant law of WePE. In particular, we do not intend to argue that the precise amount of contraction in the first layer, or the exact rate of subsequent expansion, is intrinsically fixed by the elliptic field alone. At the same time, we also do not view the observed pattern as a mere artifact of the specific dataset. Our interpretation is that it reflects a stable interaction between two factors: (i) the stronger locality-oriented geometric prior introduced by WePE, and (ii) the standard depth-wise dynamics of self-attention, in which deeper layers gradually integrate information over broader spatial extents. Under this interpretation, the *qualitative* trajectory—more compact attention in early layers followed by progressively broader attention in deeper layers—is a meaningful consequence of the inductive bias of WePE, even if the exact numerical curve can vary across datasets, architectures, and optimization settings. This interpretation is consistent with the surrounding analyses in the manuscript. The entropy analysis shows that WePE yields more concentrated and less noisy attention in early layers, while the effective attention range analysis in Figure 17 shows that this compactness is gradually relaxed with depth. Together, these results suggest that WePE encourages a local-to-global processing pattern: early layers first exploit nearby, geometrically coherent evidence, and deeper layers progressively expand the receptive field to support larger-scale integration. We therefore view the multi-scale behavior in Figure 17 as an emergent depth-wise effect of the WePE prior rather than as a dataset-specific accident. That said, we agree that the *magnitude* and *shape* of the trajectory are likely to depend on the data distribution and task demands. For example, datasets dominated by local texture cues may favor a longer persistence of compact attention, whereas tasks requiring earlier long-range reasoning may induce faster expansion. Accordingly, the safest claim is not that Figure 17 reveals a universal closed-form law, but that WePE tends to bias the model toward a *stable local-to-global multi-scale progression.*

In summary, our answer is that the observed expansion trajectory is *not* best viewed as a mere dataset artifact, but neither do we claim that its exact numerical form is an intrinsic invariant of the elliptic field. Rather, it is a stable qualitative pattern arising from the interaction between the locality bias of WePE and the standard depth progression of transformer representations.

Response to the question on Appendix D.1 and the compatibility between distance decay and periodicity.

We thank the reviewer for raising this subtle and important point. We agree that a deterministic periodic function cannot exhibit *global asymptotic decay* on an unbounded domain in the usual sense, because periodicity necessarily prevents the function value (or any induced similarity) from vanishing monotonically as the displacement tends to infinity. Therefore, Appendix D.1 should *not* be interpreted as claiming global monotone decay over the full infinite periodic plane.

What Appendix D.1 establishes is instead a *local attenuation property on the finite patch domain relevant to ViTs.* More precisely, when patch locations are restricted to a compact domain contained in a single fundamental cell and bounded away from lattice poles, the normalized similarity between positional encodings at z and $z + \delta$ satisfies a local quadratic bound of the form

$$0 \leq 1 - s(z, \delta) \leq C_D \|\delta\|^2$$

for sufficiently small δ. This means that similarity is maximized at zero displacement and decreases locally under small spatial perturbations. Thus, the theorem formalizes a *finite-domain proximity bias*: nearby patches have more similar positional encodings than slightly farther ones, which is precisely the behavior relevant for image patch interactions.

The role of periodicity is therefore different from that of global decay. Periodicity does not create a single globally decaying profile across the entire plane; rather, it *tiles* the same local geometric bias across space. In other words, the doubly periodic lattice reproduces a translation-consistent locality structure in each period cell. Within each cell, similarity is locally highest near zero displacement and attenuates under small offsets; across the full plane, this local pattern repeats due to periodicity. This is fully consistent with the fundamental properties of deterministic periodic functions.

In summary, there is no contradiction between the theory in Appendix D.1 and the basic nature of deterministic periodic functions. The theorem is about *local finite-domain attenuation*, whereas periodicity ensures that this locality bias is reproduced consistently across the 2D spatial lattice.

Response to Q5: Correction of the truncation error bound in Eq. (7).

We thank the reviewer for this careful observation. We agree that the current manuscript contains an inconsistency between the scaling stated after Eq. (6) and the bound written in Eq. (7). The correct tail behavior is of order $O(1/R_{\max})$, not $O(1/R_{\max}^2)$.

More precisely, for a two-dimensional lattice Λ, the tail sum satisfies

$$\sum_{|w|>R} |w|^{-3} \leq \frac{C_\Lambda}{R}, \tag{1}$$

for some lattice-dependent constant $C_\Lambda > 0$. This follows from comparison with the corresponding planar integral:

$$\int_R^\infty 2\pi r \cdot r^{-3}\, dr = \frac{2\pi}{R}. \tag{2}$$

Accordingly, the truncation error obeys

$$E_{\text{trunc}} = \left| \sum_{|w|>R_{\max}} T_w(z) \right| \leq \sum_{|w|>R_{\max}} \frac{2|z|}{|w|^3} \leq \frac{2C_\Lambda |z|}{R_{\max}}. \tag{3}$$

We will revise Eq. (7) accordingly and remove the incorrect $O(1/R_{\max}^2)$ statement from the manuscript.

We further note that the numerical estimate immediately following Eq. (7) in the current draft—"$R_{\max} \approx 30$, leading to an error of $E_{\text{trunc}} \approx 10^{-3}|z|$"—was derived using the incorrect bound. Using the corrected bound, we obtain

$$E_{\text{trunc}} \leq \frac{2C_\Lambda}{R_{\max}} |z|. \tag{4}$$

For a square lattice Λ with fundamental period $a = 2\omega_1$, a standard integral comparison gives

$$C_\Lambda \approx \frac{2\pi}{a^2}. \tag{5}$$

In the lemniscatic setting used in our experiments, $2\omega_1 \approx 5.244$, so $a^2 \approx 27.50$ and $C_\Lambda \approx 0.229$. Substituting $R_{\max} \approx 30$ yields

$$E_{\text{trunc}} \approx \frac{2 \times 0.229}{30} |z| \approx 1.5 \times 10^{-2} |z|, \tag{6}$$

which is approximately one order of magnitude larger than the $10^{-3}|z|$ figure reported in the current draft.

We emphasise that this correction affects only the stated theoretical upper bound and the accompanying illustrative estimate; it does not alter the implementation used in our experiments. In the inference stage, the positional field is queried through a precomputed lookup table, so the online computation no longer relies on repeated lattice summation.

As demonstrated in Appendix F, the interpolation error introduced by the 256×256 lookup table is of order $O(h^2) \approx 10^{-5}$ to 10^{-6}—far below the threshold of numerical noise in deep-network training—and this guarantee is entirely independent of the truncation-error bound in Eq. (7).

Response to the comment on Eq. (82) / the Fourier expansion terminology. We thank the reviewer for pointing out that the wording around Eq. (82) is too loose. We agree that describing this expression as a "standard Fourier series" is mathematically imprecise in the present context. Since the Weierstrass elliptic function is meromorphic and contains a second-order pole, the corresponding expansion should not be presented as an ordinary Fourier series in the L^2 sense.

To avoid overstating exactness, we will revise this part of the paper in a more conservative way. Specifically, we will no longer present Eq. (82) as a standalone exact Fourier representation of $\wp(z)$. Instead, we will describe it as a periodic surrogate motivated by the lattice structure and local singular behavior of the Weierstrass field. In the revision, we will either correct this background formula with a fully checked reference-backed expression, or remove it if it is not essential to the main empirical contribution.

Response to the missing $\frac{1}{4}$ factor in Definition B.10. We thank the reviewer for catching this normalization issue. The reviewer is correct that the coefficient in Definition B.10 is missing a factor of $\frac{1}{4}$. The reason is that the $\wp$-induced elliptic curve in our setting is written in the normalized form

$$(\wp'(z))^2 = 4\wp(z)^3 - g_2\wp(z) - g_3, \tag{7}$$

rather than in the short Weierstrass form $y^2 = x^3 + ax + b$.

Under this normalization, Bézout's theorem applied to the intersection of the line $y = m(x-x_1)+y_1$ with the cubic gives, via Vieta's formulas,

$$x_1 + x_2 + x_3 = \frac{m^2}{4}, \tag{8}$$

so the group-law expression must carry the additional factor $\frac{1}{4}$, yielding the corrected formula

$$x_3 = \frac{1}{4}\left(\frac{y_2 - y_1}{x_2 - x_1}\right)^2 - x_1 - x_2. \tag{9}$$

We will correct Definition B.10 accordingly.

We note that the addition formula actually used later in Theorem B.11 is already written in its correct normalized form:

$$\wp(z_1 + z_2) = -\wp(z_1) - \wp(z_2) + \frac{1}{4}\left(\frac{\wp'(z_1) - \wp'(z_2)}{\wp(z_1) - \wp(z_2)}\right)^2. \tag{10}$$

Summary. In summary, we agree that the current manuscript contains several mathematical inaccuracies in the appendix-level exposition:

- the truncation error bound in Eq. (7) should scale as $O(1/R_{\max})$ rather than $O(1/R_{\max}^2)$;
- the numerical estimate following Eq. (7) must be revised accordingly;
- the terminology and coefficients around Eq. (82) are stated too aggressively and need revision;
- Definition B.10 is missing a normalization factor of $\frac{1}{4}$.

We will correct all four points in the revision. These corrections affect the mathematical presentation and auxiliary derivations, but do not change the implementation used in our experiments. In particular, the inference pipeline relies on a precomputed lookup table (Appendix F), whose interpolation error is bounded at the 10^{-5}–10^{-6} level and is independent of the truncation-error analysis.

Response to Question 3 (fine-tuning implementation vs. the earlier scalar surrogate equation).

We thank the reviewer for pointing out this ambiguity. The reviewer is correct that, if the earlier draft's Eq. (9) were interpreted literally as the *full* fine-tuning positional field, then it would evaluate to a purely real scalar, in which case the imaginary channels would indeed collapse and f_2, f_4 would be zero. This is *not* how the actual fine-tuning implementation used in our experiments works.

The source of the confusion is that the original draft described the fine-tuning surrogate too coarsely. In the actual code, and now in the revised manuscript, the fine-tuning module is implemented as a *complex-valued stabilized surrogate field* together with a derivative-like auxiliary branch, so that all four channels remain active. Concretely, the revised formulation defines

$$\tilde{\wp}_{\mathrm{ft}}(z) = \big(M(z)\cos\theta(z) + C(z)\big) + i\big(M(z)\sin\theta(z) + \eta\, C(z)\big),$$

and an auxiliary branch

$$\tilde{\wp}'_{\mathrm{ft}}(z) = \big(M'(z)\cos\theta(z) + C'(z)\big) + i\big(M'(z)\sin\theta(z) + \eta' C'(z)\big),$$

from which the four features are constructed as

$$f_1 = \tanh\!\big(\alpha\, \Re(\tilde{\wp}_{\mathrm{ft}}(z))\big), \quad f_2 = \tanh\!\big(\alpha\, \Im(\tilde{\wp}_{\mathrm{ft}}(z))\big),$$
$$f_3 = \tanh\!\big(\alpha\, \Re(\tilde{\wp}'_{\mathrm{ft}}(z))\big), \quad f_4 = \tanh\!\big(\alpha\, \Im(\tilde{\wp}'_{\mathrm{ft}}(z))\big).$$

Therefore, in the implementation used for the experiments, f_2 and f_4 do *not* remain zero. They are generated by the explicitly constructed imaginary branches of the stabilized surrogate and its auxiliary derivative-like field. More specifically, the imaginary activation is not an arbitrary coding trick: it is structurally induced by the polar-angle term

$$\theta(z) = \operatorname{atan2}(\Im z, \Re z).$$

For generic patch locations under the normalized coordinate mapping used in practice, $\Im(z)$ is generally nonzero, and hence $\sin\theta(z)$ is generally nonzero as well. As a result, the term $M(z)\sin\theta(z)$ provides a direct non-degenerate source for $\Im(\tilde{\wp}_{\mathrm{ft}}(z))$, while the $\eta\, C(z)$ term provides an additional periodic correction. The same mechanism applies to the auxiliary branch through $M'(z)\sin\theta(z)$ and $\eta' C'(z)$. Thus, the imaginary channels are structurally active in the fine-tuning module rather than being identically zero by construction.

To answer the reviewer's question directly: the codebase does *not* use a purely real scalar fine-tuning field with vanishing imaginary channels. Rather, the earlier equation in the draft was an oversimplified intermediate description that did not faithfully expose the full complex-valued implementation. More precisely, the original intention of the earlier Eq. (9) was only to illustrate the dominant *radial* surrogate term in a simplified, motivational form. It was not intended to be read as the final implemented 4-channel module. The revised manuscript corrects this by presenting the complete fine-tuning construction explicitly, including the coupled real/imaginary channels and the derivative-like auxiliary branch.

In summary, the reviewer's concern is valid for the *earlier wording*, but it does not apply to the actual implementation used in our experiments. The apparent discrepancy arose from an incomplete presentation in the draft, which is precisely why we expanded and corrected the fine-tuning description in the revised manuscript.